



Try it now!

IEP IEP Planner

Quiz Creator
Generate a quiz to test your students

Lesson Planner
Create a standards-aligned lesson plan for your students.

Assign to class

Class Analytics



Last Modified: May 01, 2026

What you will learn

- Movement of molecules across the cell membrane
- Diffusion and facilitated diffusion processes



[Figure 1]



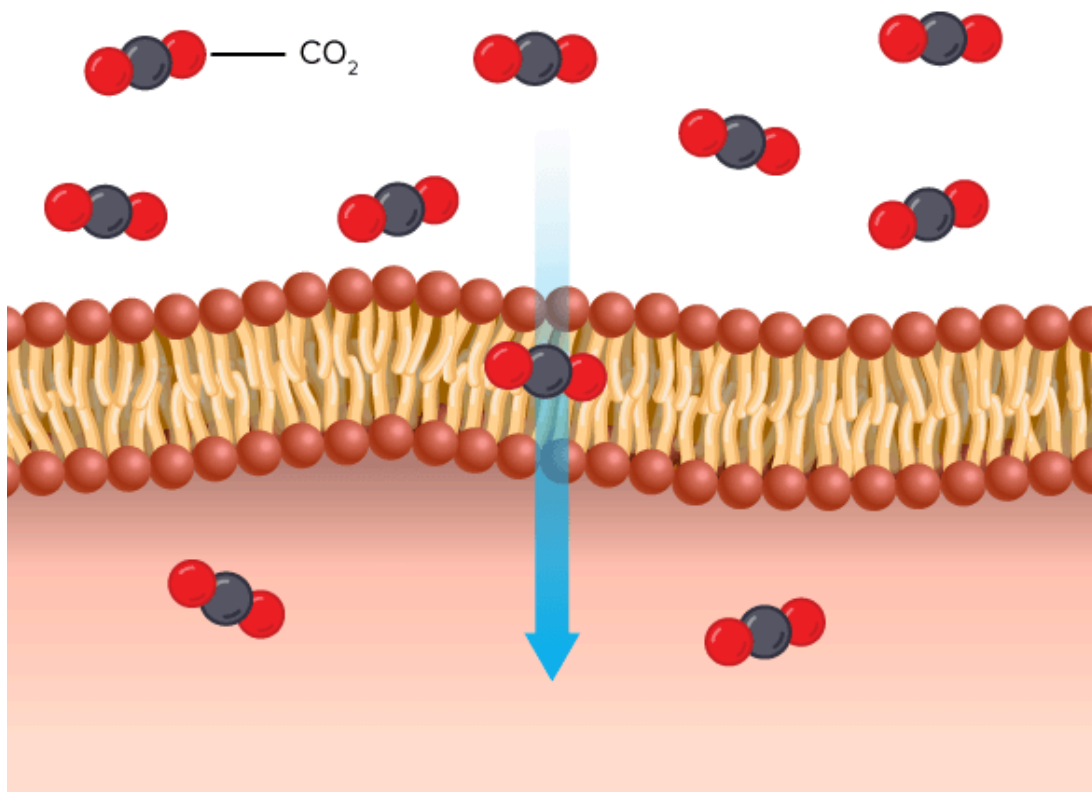
Often a large truck might be needed if you are moving large boxes and furniture. Just as a truck is needed to help move these items from one location to another, a **cell** needs help moving things as well, or facilitating the diffusion process. This would be the job of a special type of **protein**.

Diffusion and Facilitated Diffusion

Diffusion

Diffusion is the movement of molecules from an area of high concentration of the molecules to an area with a lower concentration. The difference in the concentrations of the molecules in the two areas is called the **concentration gradient**. Diffusion will continue until this gradient has been eliminated. Since diffusion moves materials from an area of higher concentration to the lower, it is described as moving solutes "down the concentration gradient." The end result of diffusion is an equal concentration, or **equilibrium**, of molecules on both sides of the membrane.

If a molecule can pass freely through a cell membrane, it will cross the membrane by diffusion (see the **Figure** below).



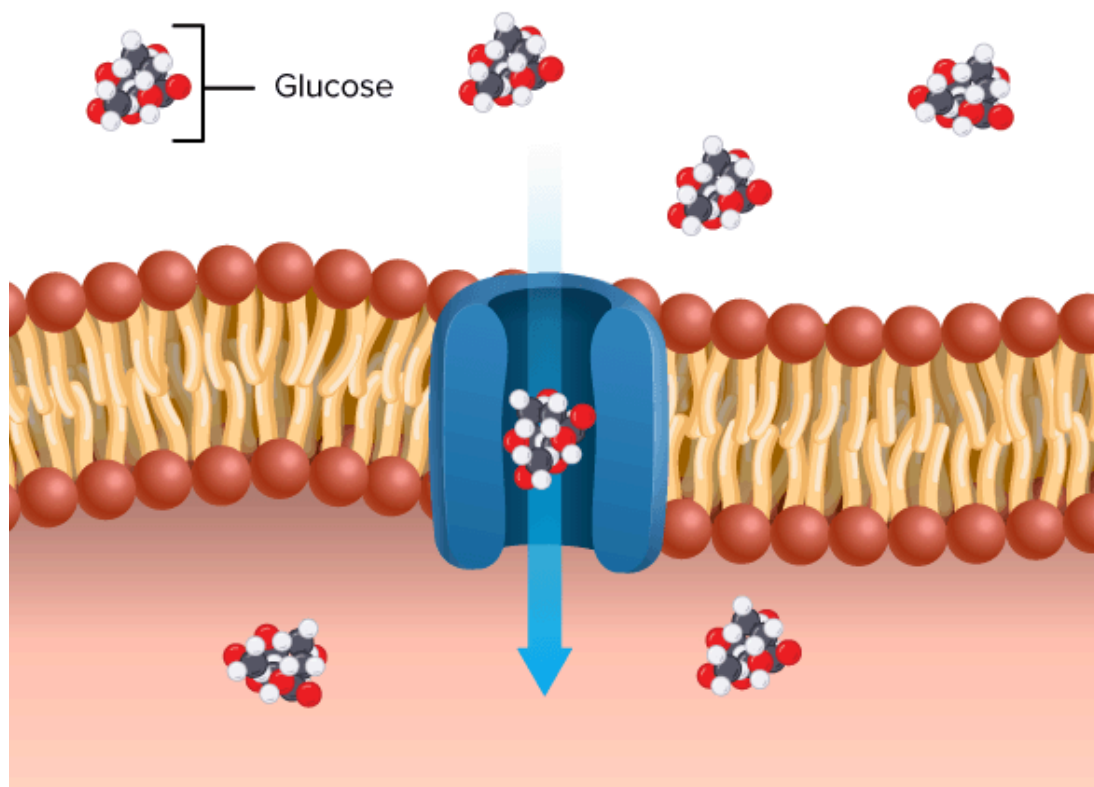
[Figure 2]

Molecules move from an area of high concentration to an area of lower concentration until an equilibrium is met. The molecules continue to cross the membrane at equilibrium, but at equal rates in both directions.



Facilitated Diffusion

What happens if a substance needs assistance to move across or through the cell membrane? **Facilitated diffusion** is the diffusion of solutes through **transport proteins** in the cell membrane. Facilitated diffusion is a type of **passive transport**. Even though facilitated diffusion involves transport proteins, it is still passive transport because the solute is moving down the concentration gradient (see the **Figure below**).



[Figure 3]

Facilitated Diffusion



Nonpolar molecules can easily diffuse across the cell membrane. However, due to the hydrophobic nature of the lipids that make up cell membranes, polar molecules (such as

or ions to diffuse across the membrane. **Channel proteins**, **gated channel proteins**, and **carrier proteins** are three types of transport proteins that are involved in facilitated diffusion.

A *channel protein*, a type of transport protein, acts like a pore in the membrane that lets **water** molecules or small ions through quickly. Water channel proteins (aquaporins) allow water to diffuse across the membrane at a very fast rate. **Ion channel** proteins allow ions to diffuse across the membrane.

A *gated channel protein* is a transport protein that opens a "gate," allowing a molecule to pass through the membrane. Gated channels have a binding site that is specific for a given molecule or ion. A **stimulus** causes the "gate" to open or shut. The stimulus may be chemical or electrical signals, temperature, or mechanical force, depending on the type of gated channel. For example, the sodium gated channels of a **nerve** cell are stimulated by a chemical signal which causes them to open and allow sodium ions into the cell.

A *carrier protein* is a transport protein that is specific for an ion, molecule, or group of substances. Carrier proteins "carry" the ion or molecule across the membrane by changing shape after the binding of the ion or molecule. Carrier proteins are involved in passive and **active transport**. **Glucose** molecules are too big to diffuse through the cell membrane easily, so they are moved across the membrane through carrier proteins. In this way glucose diffuses very quickly across a cell membrane, which is important because many cells depend on glucose for **energy**.



Facilitated diffusion through the cell membrane. Channel proteins and carrier proteins are shown. Water molecules and ions move through channel proteins. Glucose molecules are carried across the cell membrane by carrier proteins.

Ion Channels

Ions such as sodium (Na^+), potassium (K^+), **calcium** (Ca^{2+}), and chloride (Cl^-), are important for many cell functions. Because they are charged (polar), these ions do not diffuse through the membrane. Instead they move through ion channel proteins where they are protected from the hydrophobic interior of the membrane. *Ion channels* allow the formation of a

always open, others are "gated" and can be opened or closed. Gated ion channels can open or close in response to different types of stimuli, such as electrical or chemical signals.



DID YOU KNOW?

Voltage-gated ion channels located in **neurons** are good examples of transport proteins. These ion channels open in response to changes in the cell membrane's electrochemical potential and allow neurons to fire a **nerve impulse**.

Check your understanding!

- Facilitated diffusion is the diffusion of solutes through transport proteins in the cell membrane. Channel proteins, gated channel proteins, and carrier proteins are three types of transport proteins that are involved in facilitated diffusion.
-

Review

1. Explain how diffusion and facilitated diffusion differ.
2. What is facilitated diffusion?
3. What is a transport protein? Give three examples.
4. Assume a molecule must cross the cell membrane into a cell. The molecule is very large. How will it be transported into the cell?
5. Explain how carrier proteins function?
6. Explain the role of ion channels. Why are ion channels necessary?



Image Attributions



Report Content Errors



[Our mission](#)
[Meet the team](#)
[Partners](#)
[Press](#)
[Efficacy studies](#)
[Careers](#)
[Security](#)
[Blog](#)
[CK-12 usage map](#)
[Testimonials](#)

[Certified Educator Program](#)
[CK-12 trainers](#)
[Webinars](#)
[CK-12 resources](#)
[Help](#)
[Contact us](#)

[Common Core Math](#)
[K-12 FlexBooks](#)
[College FlexBooks](#)
[Tools and apps](#)



v2.11.14.20260514061705-33eb675369

© CK-12 Foundation 2026 | FlexBook Platform®, FlexBook®, FlexLet® and FlexCard™ are registered trademarks of CK-12 Foundation.

[Terms of use](#)

[Privacy](#)

[Attribution guide](#)

